

Chapter 9

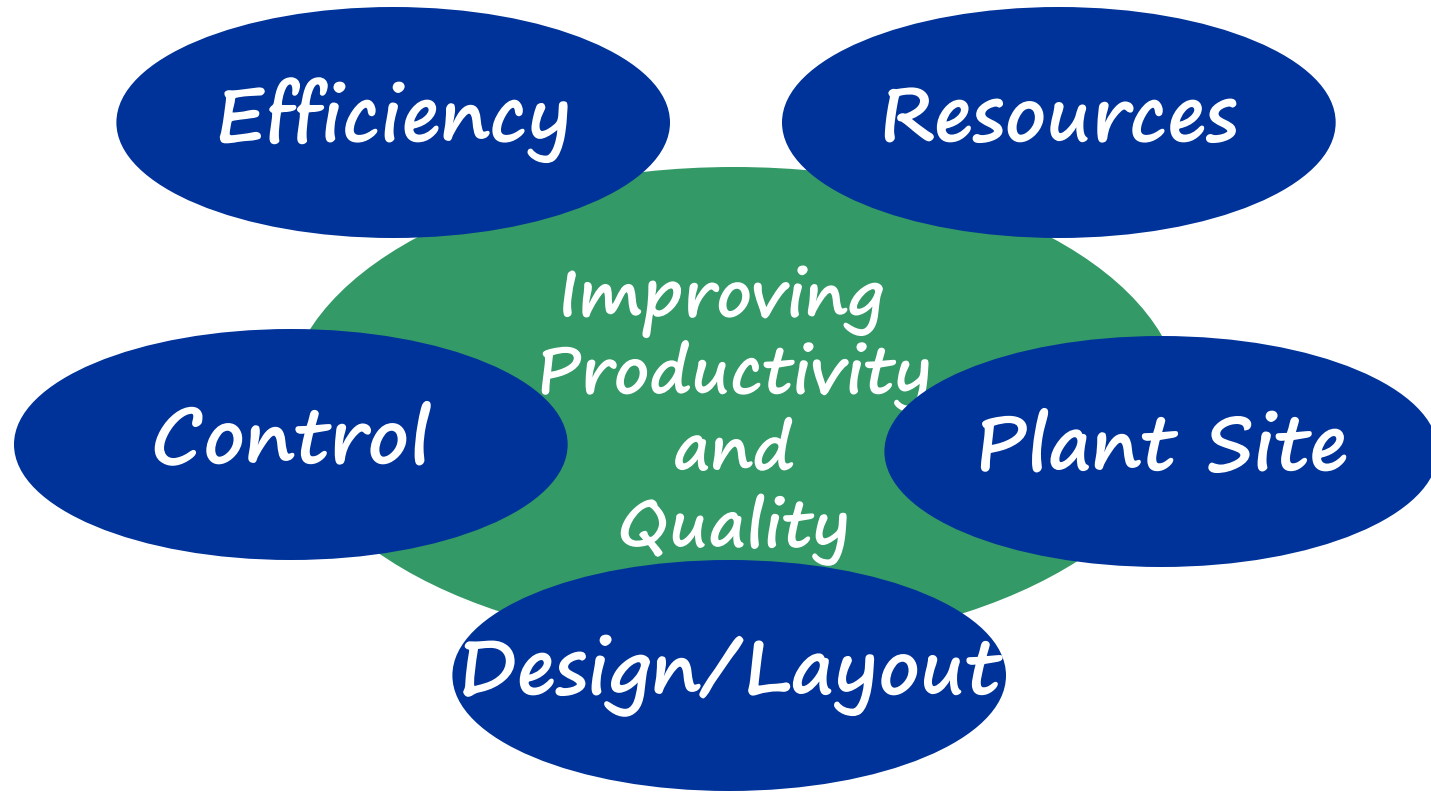


INTRODUCTION
TO BUSINESS 4e

JEFF MADURA

Improving Productivity and Quality

Learning Objectives

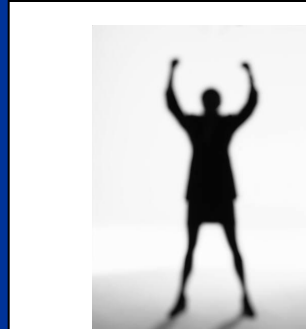


Improving
Productivity
and Quality

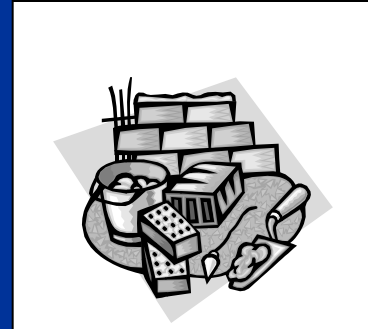
Resources
Plant Site
Design/Layout
Control
Efficiency



Resources Used



H.R.



Materials



Other

Work Station
Assembly Line

Improving
Productivity
and Quality

Resources
Plant Site
Design/Layout
Control
Efficiency



Site Selection

COST

Workplace Space

Labor

ACCESS

Tax Incentives

Transportation

SOURCES

Demand

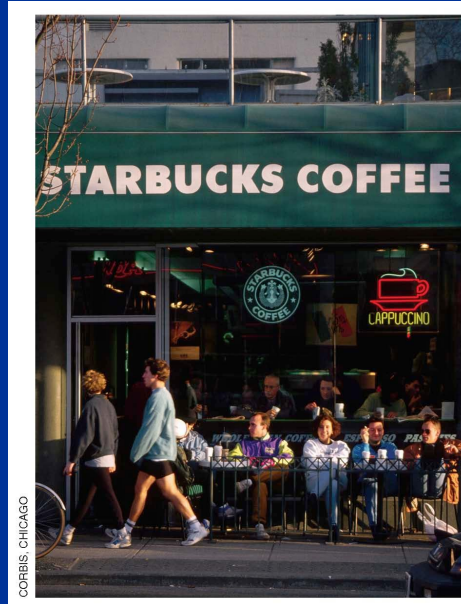
Labor

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Resources
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Evaluating Sites



Identify

Weigh

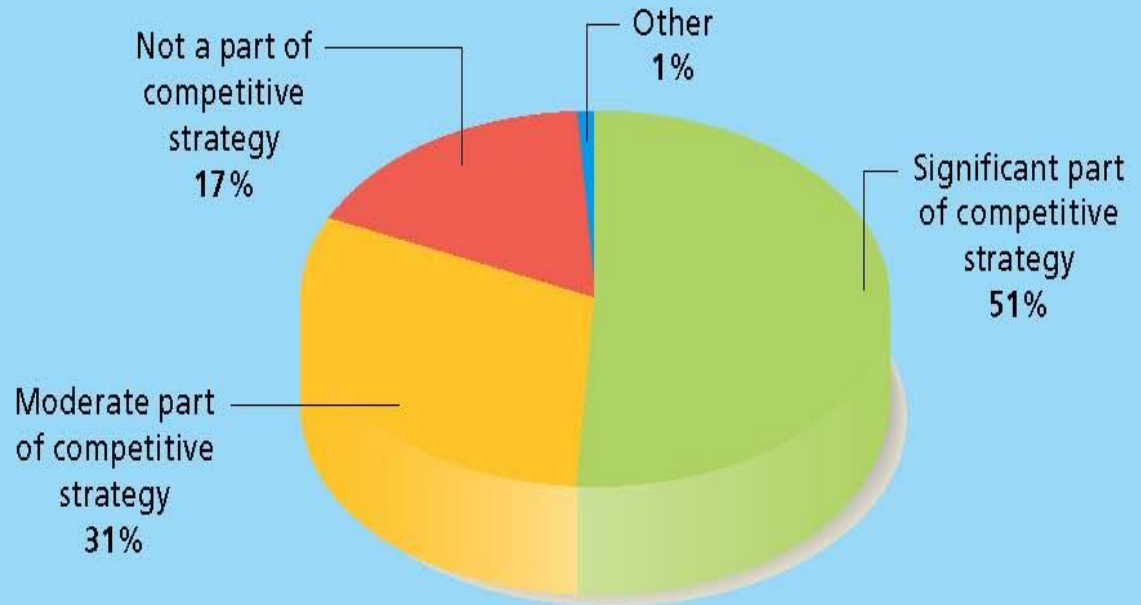
Evaluate

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Location Investment

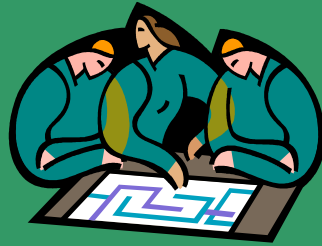


Improving
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Resources
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Design & Layout



Site Characteristics

Production

Product Layout

Fixed Position

Flexible Manufacturing

Improving
Productivity
and Quality

Resources
Plant Site
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Efficiency



Plan for Growth



Production Capacity

Hotelling (JIT)

Improving
Productivity
and Quality

Resources
Plant Site
Design/Layout
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Efficiency



Key Tasks

Production control



Purchasing

Inventory

Routing

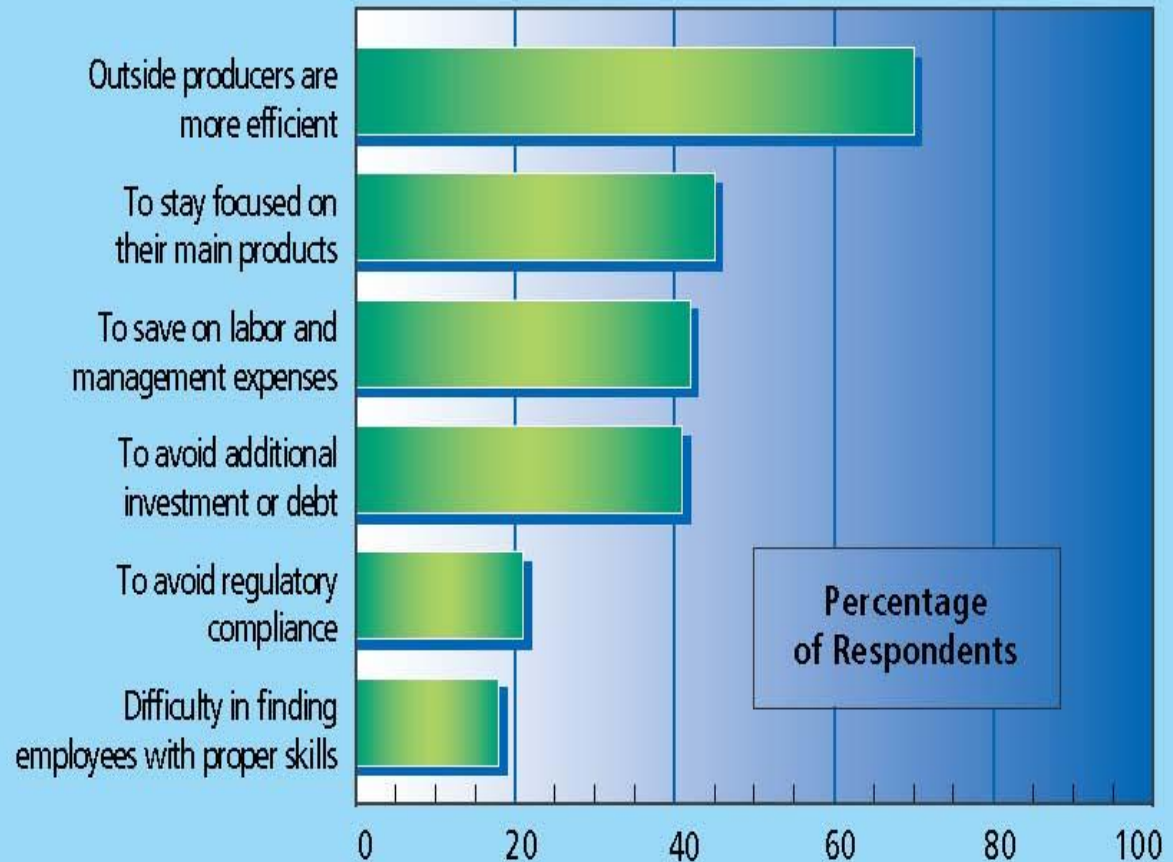
Scheduling

Quality

Improving
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and Quality

Resources
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Efficiency

Purchasing Materials

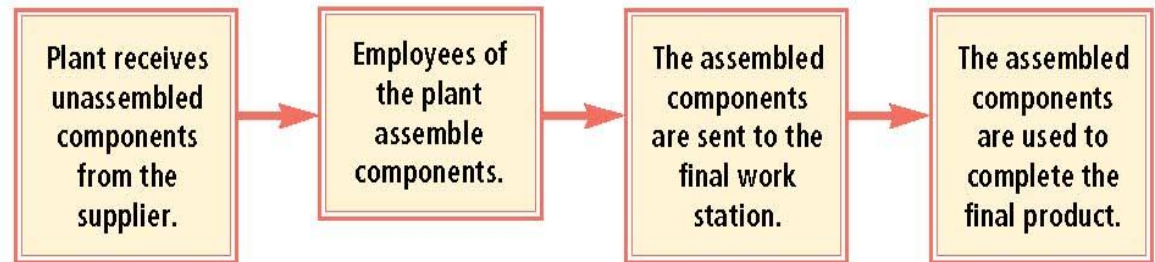


Improving
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Resources
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Deintegration

Previous Production Process



Revised Production Process



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Deintegration



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Deintegration



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Deintegration



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Inventory Control



Materials

Work-in-Process

Finished Goods

Improving
Productivity
and Quality

Inventory Control



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Routing

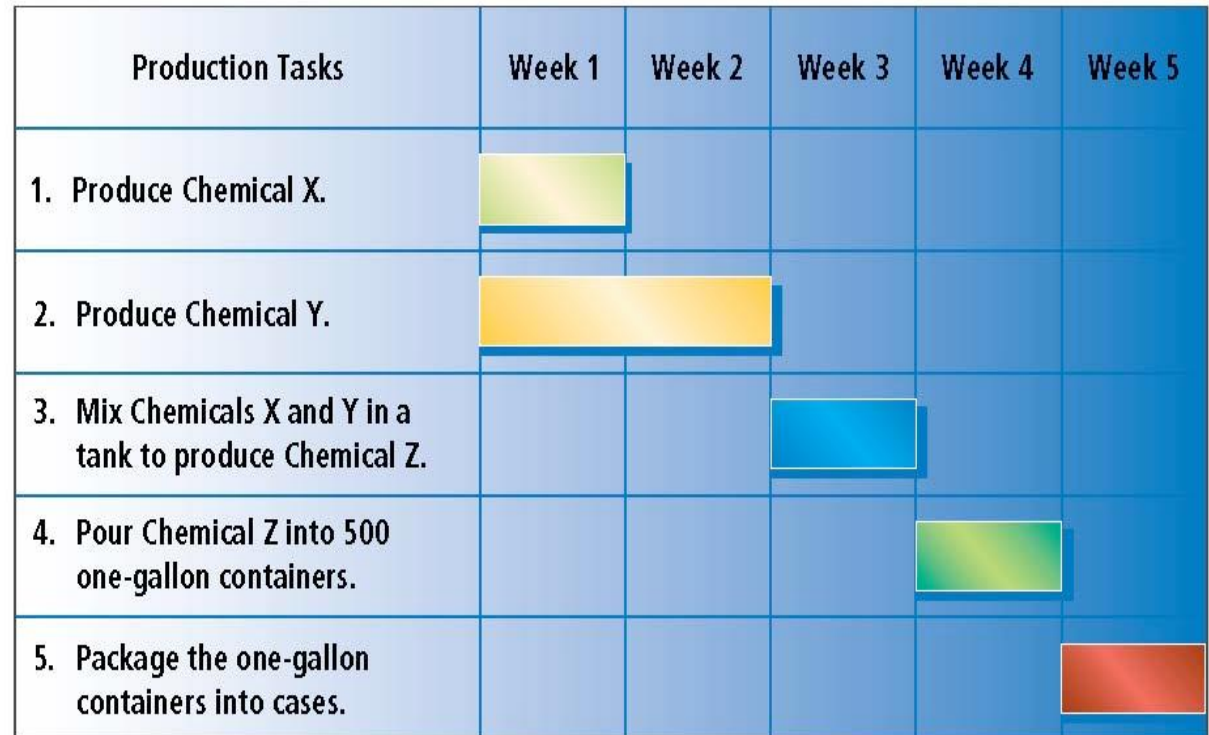


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Resources
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Production Scheduling

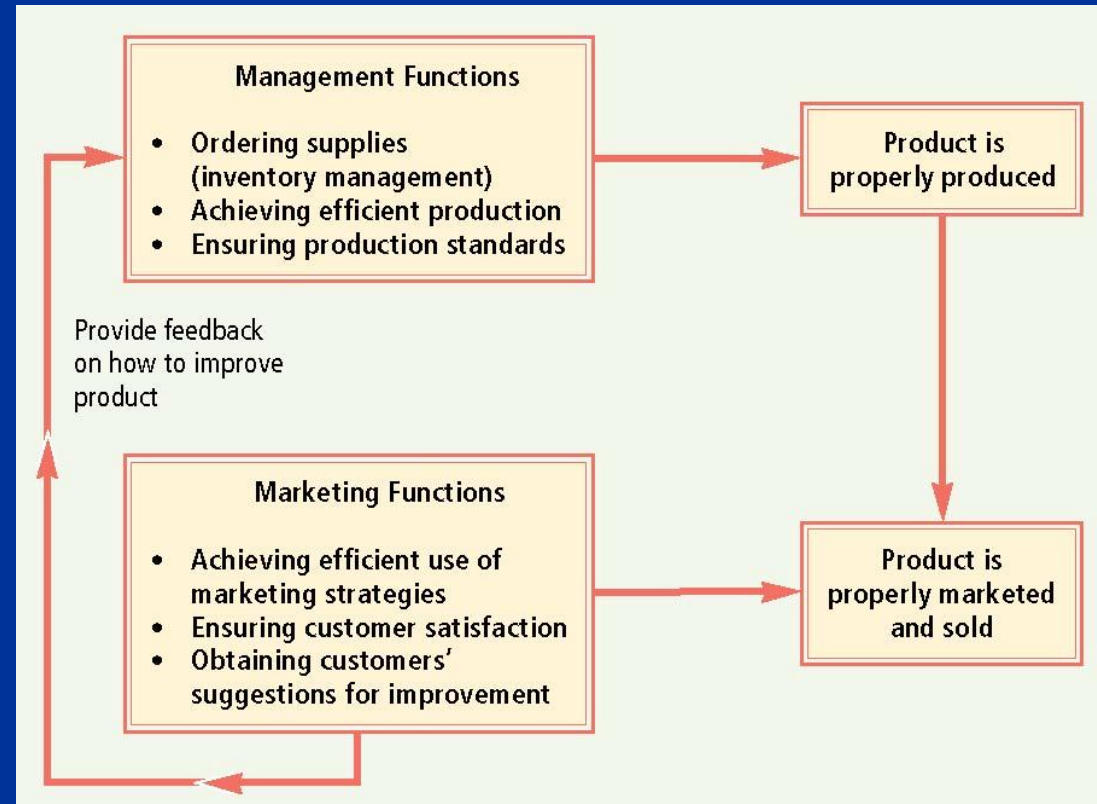
Gantt Chart



Improving Productivity and Quality

Resources
Plant Site
Design/Layout
Control
Efficiency

Quality



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Efficiency



Quality Control



Technology

Employees

Sampling

Complaints

Improving
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Resources
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Efficiency



Quality



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Improving
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Production Efficiency



Efficiency

Benchmarking

Stretch Targets

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Efficiency Methods



Technology

Economies of Scale

Restructuring

Review & Critical Thinking Questions

What general factors influence design and layout decisions?



Review & Critical Thinking Questions

The following factors influence design and layout decisions: (1) site characteristics, (2) production processes, (3) product lines, and (4) desired production capacity.



Review & Critical Thinking Questions

What is outsourcing? What are the advantages and disadvantages of outsourcing?



Review & Critical Thinking Questions

Outsourcing involves the purchase of parts from suppliers rather than producing the parts. Outsourcing can reduce a firm's expenses if suppliers can produce the parts at a lower cost than the manufacturer. It may also be better to have the supplier partially assemble the components before sending them to the manufacturer; some of the assembly task is thereby shifted to the supplier. Although outsourcing can be beneficial, it places much responsibility on another manufacturing company. Thus, the ability of a firm that outsources to meet its production schedule depends on the manufacturing companies that it relies on.



Review & Critical Thinking Questions

Compare just-in-time (JIT) inventory with materials requirements planning (MRP).



Review & Critical Thinking Questions



The just-in-time system attempts to reduce materials inventories to a bare minimum by frequently ordering small amounts of materials. The goal of just-in-time inventory management is to reduce the costs of holding inventories. However, there is a cost of managerial time required for frequent ordering and a cost of frequent deliveries. In addition, the just-in-time system could result in a shortage if needs are inaccurately forecasted.

Review & Critical Thinking Questions

Materials requirements planning (MRP) is a process for ensuring that the raw materials are available when needed. It normally requires the use of a computer and helps managers determine the amount of specific materials that should be acquired at any given time. As the firm forecasts the demand for its product in the future, it can determine the time at which the materials need to arrive to achieve a production level that will accommodate the forecasted demand.

Both JIT and MRP are part of an efficient production management system.

